



TCC805x MCU BSP

API Specification

for Analog Digital Converter Controller

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the analog digital converter controller device driver (hereinafter referred to as ADC driver) of TCC805x MCU BSP. For more detailed information on the analog digital converter controller in the TCC805x MCU Sub-system, refer to Chapter 17 Analog Digital Converter (ADC) Controller of the “*PART 11-MICOM SUB-SYSTEM*” in the “*TCC805x Full Specification*”. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

ADC	Analog Digital Converter
-----	--------------------------

3 SOURCE FILES

3.1 Location of Source Files

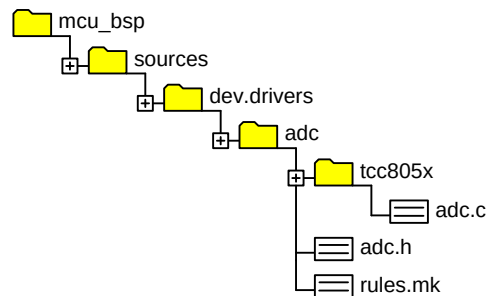


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `adc.h`
 - API header of ADC driver
- `adc.c`
 - Source of ADC driver
- `rules.mk`
 - Make rules

4 ENUMERATIONS

4.1 ADCChannel_t

Enumeration variables for ADC channel value of ADC user interface API. TCC805x ADC supports 12 channels.

Declaration

```
typedef enum __ADC_CHANNEL__  
{  
    ADC_CHANNEL_0      = 0,  
    ADC_CHANNEL_1      = 1,  
    ADC_CHANNEL_2      = 2,  
    ADC_CHANNEL_3      = 3,  
    ADC_CHANNEL_4      = 4,  
    ADC_CHANNEL_5      = 5,  
    ADC_CHANNEL_6      = 6,  
    ADC_CHANNEL_7      = 7,  
    ADC_CHANNEL_8      = 8,  
    ADC_CHANNEL_9      = 9,  
    ADC_CHANNEL_10     = 10,  
    ADC_CHANNEL_11     = 11,  
    ADC_CHANNEL_MAX     = 12  
} ADCChannel_t;
```

Description

Value	Description
ADC_CHANNEL_xx	Channel number value for each ADC channel
ADC_CHANNEL_MAX	The Maximum number of channels

Remark

None

4.2 ADCMode_t

Enumeration variables for selecting ADC mode

Declaration

```
typedef enum __ADC_MODE__  
{  
    ADC_MODE_NORMAL    = 0,  
    ADC_MODE_IRQ       = 1,  
    ADC_MODE_DMA        = 2  
} ADCMode_t;
```

Description

Value	Description
ADC_MODE_NORMAL	The conversion value can be read in the ADC sampled data register (SD_OUT0 ~ SD_OUT11).
ADC_MODE_IRQ	IRQ interrupt is generated when the conversion operation is completed.
ADC_MODE_DMA	The conversion value can be read in the designated location of the DMA.

Remark

None

5 CONSTANTS

5.1 Base Address and Offset

Constant values for TCC805x ADC base address and register offset

Declaration

```
#define ADC_BASE_ADDR      (0x1B700000UL)
#define ADC_SMP_CMD_STS_REG 0x00
#define ADC_IRQ_DREQ_STS_REG 0x01
#define ADC_CTRL_CFG_REG   0x02
#define ADC_SDOUT0_REG      0x20
#define ADC_SDOUT1_REG      0x21
#define ADC_SDOUT2_REG      0x22
#define ADC_SDOUT3_REG      0x23
#define ADC_SDOUT4_REG      0x24
#define ADC_SDOUT5_REG      0x25
#define ADC_SDOUT6_REG      0x26
#define ADC_SDOUT7_REG      0x27
#define ADC_SDOUT8_REG      0x28
#define ADC_SDOUT9_REG      0x29
#define ADC_SDOUT10_REG     0x2A
#define ADC_SDOUT11_REG     0x2B
```

Description

Value	Description
ADC_BASE_ADDR	ADC Register base address
ADC_SMP_CMD_STS_REG	ADC Sampling Command and Control Status Register
ADC_IRQ_DREQ_STS_REG	ADC Controller Interrupt and DMA Request Status Register
ADC_CTRL_CFG_REG	ADC Control Operation Configuration Register
ADC_SDOUT0_REG	1st Sampled Data Register
ADC_SDOUT1_REG	2nd Sampled Data Register
ADC_SDOUT2_REG	3rd Sampled Data Register
ADC_SDOUT3_REG	4th Sampled Data Register
ADC_SDOUT4_REG	5th Sampled Data Register
ADC_SDOUT5_REG	6th Sampled Data Register
ADC_SDOUT6_REG	7th Sampled Data Register
ADC_SDOUT7_REG	8th Sampled Data Register
ADC_SDOUT8_REG	9th Sampled Data Register

<i>Value</i>	<i>Description</i>
<i>ADC_SDOUT9_REG</i>	<i>10th Sampled Data Register</i>
<i>ADC_SDOUT10_REG</i>	<i>11th Sampled Data Register</i>
<i>ADC_SDOUT11_REG</i>	<i>12th Sampled Data Register</i>

Remark

None

6 APIs

6.1 ADC_Init

Initializes the ADC. Regardless of the input type, the **ADC_Init()** function performs ADC clock enable.

There are three types of mode in ADC as follows:

- Normal mode: Default mode
- IRQ mode: When conversion is completed, this mode can generate an interrupt.
- DMA mode: By using DMA, this mode can read the conversion result.

Declaration

```
void ADC_Init
(
    ADCMode_t  iType
);
```

Parameters

Value	Description
iType	[in] ADC Mode (Normal, IRQ, DMA). Refer to the ADCMode_t .

Return

None

Remark

None

6.2 ADC_Read

Starts the ADC conversion using the **ADC_SMP_CMD** register for the set input channel, checks the conversion done bit, and reads the conversion value from the **SDOUT** register.

Declaration

```
uint32 ADC_Read
(
    ADCChannel_t  iChannel
);
```

Parameters

Value	Description
<i>iChannel</i>	<i>[in] ADC Channel number. Refer to the ADCCchannel t.</i>

Return

Value	Description
<i>uint32</i>	<i>ADC conversion value, conversion value range is 0 ~ 4096(12 bits).</i>

Remark

None

6.3 ADC_AutoScan

This function calls **ADC_Read()** function internally, to perform conversion for all ADC channels and to read the conversion values.

Declaration

```
uint32 * ADC_AutoScan  
(  
    void  
);
```

Parameters

None

Return

Value	Description
<i>uint32 *</i>	<i>Static array address storing all channel conversion values</i>

Remark

None

7 HOW TO USE ADC DRIVER

7.1 Initialize and Read Data

```
int main()
{
    uint32 *dev_read_buf = NULL;
    uint32 read_data      = 0;

    ADC_Init(ADC_MODE_NORMAL);           //ADC Normal Mode initialization
    read_data = ADC_Read(ADC_CHANNEL0); //Read conversion value for ADC channel 0
    dev_read_buf = ADC_AutoScan();       //Read conversion value for ALL ADC channels 0~11

    Return 0;
}
```

8 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release

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